S2 E15

BK

Season 2: Retrieval and RAG

When RAG is the wrong answer.

RAG answers a question by finding a passage. Some answers are not passages. Four alternatives, what this season measured, and how to choose.

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Applied AI

Why it matters

Not every answer is sitting in a document

Fourteen episodes built a system that finds the right passage and answers from it. It now gets 354 of 400 right. But look at what people ask:

A passage
RAG's home ground

The answer is written down, in one place.

`"How long is the warranty on the Brio kettle?"`

A join
two facts, linked

The answer combines two records.

`"How quickly are urgent safety visits made to my home in postcode area B?"`

A count
a calculation

The answer is computed, not written anywhere.

`"How many kettles were returned last month?"`

A live state
true right now

The answer changes by the minute.

`"Where is my order?"`

The first two are from the season's test set. The last two are illustrations: the test set has none.

Recap: what RAG is built to do

1	2	3	4	5	6	7	8	9	10	11
Load	Chunk	Embed	Rewrite	**Search**	Filter	**Rerank**	**Prompt**	**Answer**	Cite	Score

Find
search, filter and rerank pick the best 5 passages

→

Read
the model reads those 5, and nothing else

→

Answer
in its own words, citing each passage

Good at

Answers written down somewhere in a large, changing library, scoped to who is asking.

Not built for

Answers that must be computed, that live in a database or a live system, or that are about how the model should behave.

The question for this episode

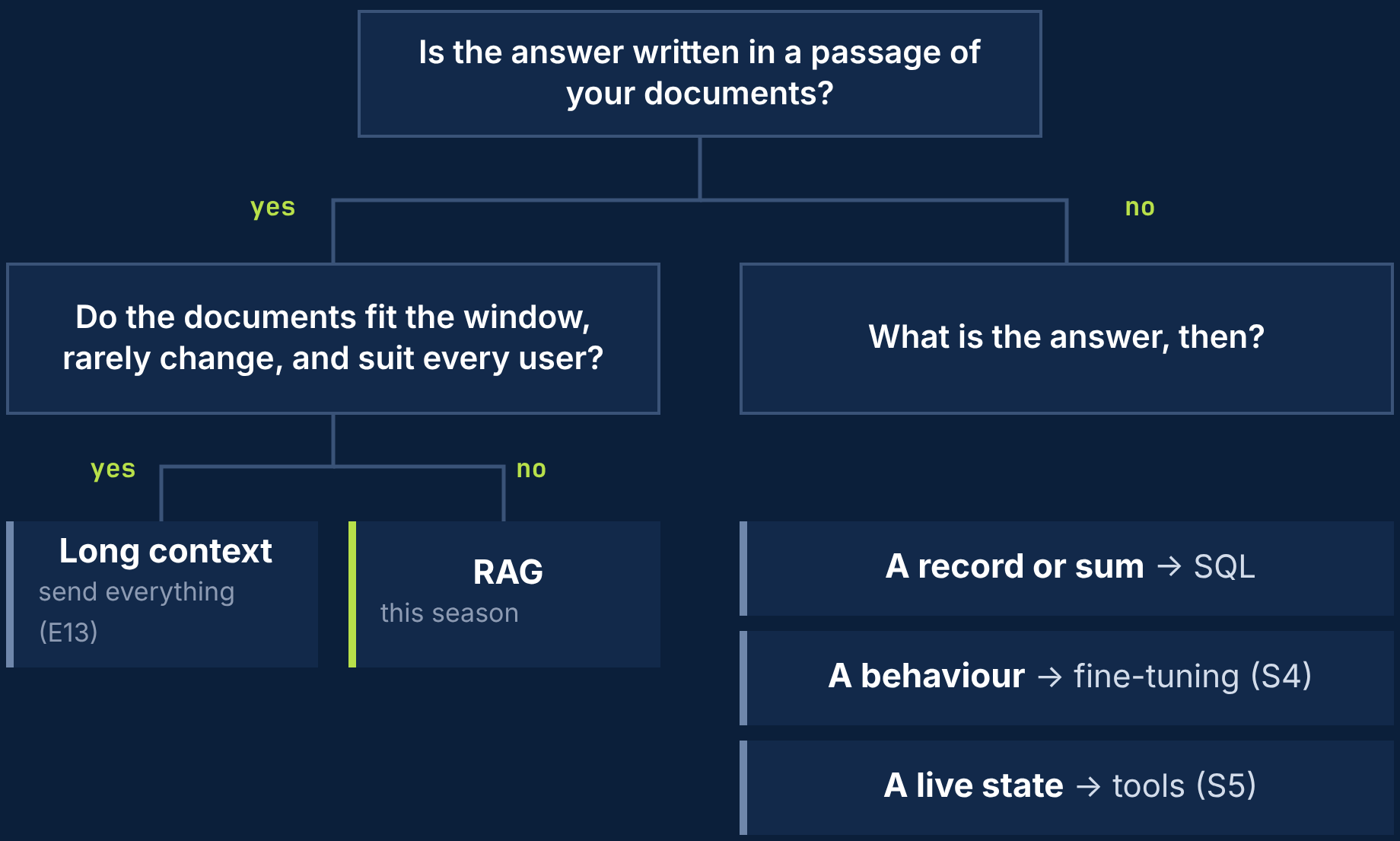
Before building RAG, ask: is the answer a passage? If not, what is it?

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Mental model

Mental model: is the answer a passage?

Two questions decide most cases. Ask them before you build anything.



One case no technique fixes

If your documents disagree with each other, every option inherits the conflict (slide 11).

Option 1

Option 1, long context: just send everything

Measured in E13. Halvard's whole public library, 160,000 tokens, sent with every question.

	LONG CONTEXT	RETRIEVAL
Hard questions right, of 80	**75**	**71**
Simple questions right, of 40	**39**	**39**
Seconds median per question	**1.9**	**2.5**
Cost per question	**$0.016**	**$0.0016**

The verdict About as accurate, no slower, no pipeline to build. Ten times the price.

Option 1

When long context stops working

E13 handed it a clean library: only current, public documents. A real one is rarely that tidy.

Who may see what	Staff pages must never reach a customer. Send everything, and you need a separate library per kind of user.
It changes	Old versions must be removed before every question. A user’s pasted copy may already be out of date.
It grows	The price grows with every document, on every question. Past about six times Halvard’s library, it no longer fits.

Use it when

The documents fit with room to spare, change rarely, and every user may see all of them.

Option 2

Option 2, structured data: look the answer up exactly

A bridge question joins two facts. Search must find both passages at once; a table needs one lookup.

A BRIDGE QUESTION FROM THE TEST

How quickly are urgent safety visits made to my home in postcode area B?

Answer: 32 hours

FOUND BY RETRIEVAL

14 of 20

bridge questions answered right (E13).

STORED AS A TABLE

```
postcode B → Midlands
Midlands → 32 hours
```

One exact join, every time.

Use it when	The answer is a lookup or a calculation over records: prices, regions, counts, totals. Store the records, and query them.

Reasoning, not a result: the season did not build or test a database.

Option 3

Option 3, fine-tuning: change behaviour

Retrieval adds facts to the prompt. It does not change how the model writes, reasons or formats.

RAG SUPPLIES

What to say

The warranty period, the error code, the policy wording. Facts that change and must be current.

FINE-TUNING SHAPES

How to say it

Your tone of voice, a fixed output format, a specialised way of reasoning. Behaviour that stays the same.

A common mistake

Fine-tuning a model to teach it facts. The facts go stale with the next policy change, and the model cannot cite where they came from.

Reasoning, not a result: Season 4, Adapting, covers fine-tuning in full.

Option 4

Option 4, tools: ask the live system

"Where is my order?" is not written in any document. It is the state of a system, right now.

The question
"Where is my order?"

→

A tool call
the model asks the order system for its status

→

The answer
from live data, current to the second

Why not RAG	An index is a snapshot. By the time a status is indexed, it may already be wrong.
Use it when	The answer is true right now: an order, a balance, an appointment, a stock level.

Reasoning, not a result: Season 5, Agents, covers tools in full.

Mental model

Real assistants combine them

One customer conversation can need three of the four. The skill is routing each question to the right one.

"Is my kettle under warranty?"	**A record:** the purchase date, from the orders database.
"What does the warranty cover?"	**A passage:** the warranty policy, by RAG.
"When will the engineer arrive?"	**A live state:** the booking system, by a tool call.

So the pattern is

RAG is one tool in the kit, not the whole kit. Choose by the shape of each answer, not by what you built last.

An illustration, not from the test set.

Where it breaks

When no technique helps: the documents disagree

In E14, 3 of the final system's 5 failures with the right passage in the prompt had the same cause.

ONE OF THE THREE

How long is the warranty on the Halvard Home Mira robot vacuum from the date of purchase?

A SUPPORT ARTICLE SAYS	THE WARRANTY POLICY SAYS
"a two-year warranty from purchase"	"40 months from delivery"

Every option fails	RAG, long context, SQL or a fine-tuned model: each inherits the conflict, or hides it by picking one.

The fix

The content, not the pipeline. Decide which document is right, and correct the other.

Where it breaks

What this season measured, and what it did not

Only one of the four alternatives was tested. Treat the rest as reasoning to check on your own questions.

Long context	E13: same accuracy within luck, ten times the price, on 120 questions.	measured
Documents disagree	E14: 3 of the 5 in-prompt failures.	measured
SQL	Bridge questions suggest it; no database was built.	reasoned
Fine-tuning	Not run. Season 4.	reasoned
Tools	Not run. Season 5.	reasoned

The decision

Match the tool to the shape of the answer

One table to keep.

A passage in a large or changing library	**RAG** `this season`
A passage in a small, stable library	**Long context** `E13`
A record or a sum lookups and calculations	**SQL** `structured data`
A behaviour tone, format, reasoning	**Fine-tuning** `Season 4`
A live state true right now	**Tools** `Season 5`

For engineers

Run the evidence yourself

This episode runs nothing new. Its evidence comes from E13 and E14.

```
# E13: long context against retrieval, no key, no cost
$ uv run python field-notes/s2-e13-long-context-or-retrieval/analyse.py

# E14: trace every wrong answer, no key, no cost
$ uv run python episodes/s2-e14-tracing/analyse.py
```

Cost	Nothing. Both read committed results.

For leaders

Know what your users actually ask

The right tool depends on the shape of the answers. Most teams have never sorted their questions that way.

Ask your team

What share of our questions are lookups, calculations or live state, not passages?

Ask your team

Did we choose RAG because it fits our questions, or because it was the pattern we knew?

Ask your team

Who owns fixing documents that contradict each other?

Decision rule

Use RAG when the answer is a passage in your documents; use something else when it is a calculation, a behaviour, or the state of a live system.

Next: S2 E16, the Season 2 recap

Sixteen episodes, one system, from 182 wrong answers to 46. What each stage taught, in one place.

Where this fits

The series map

S1 LLMs	**S2** Retrieval	**S3** Strategy
S4 Adapting	**S5** Agents	**S6** Economics
S7 Data	**S8** Evaluation	**S9** AI teams
S10 Reliability	**S11** Security	**S12** Delivery
S13 Performance	**S14** Resilience	**S15** Governance

☐ Leadership season

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Episode guide brendankellyai.github.io

Code github.com/BrendanKellyAI/applied-ai-lab